INTERMOLECULAR FORCES

OBJECTIVES

- To measure temperature changes caused by the evaporation of several liquids.
- To relate the observed temperature changes to the strength of intermolecular forces of attraction present in each liquid.

INTRODUCTION

In this experiment, temperature probes will be placed in various pure liquids. Evaporation—which occurs when the probe is removed from the liquid's container—is an endothermic process that results in a temperature decrease as the temperature probe provides some of the heat needed for evaporation. The magnitude of this temperature decrease is related to the strength of intermolecular forces of attraction among the molecules evaporating.

Initially, you will investigate the molecular structures of these molecules and predict the relative (to ethanol) temperature changes you will observe. Remember that you are *indirectly* assessing the strength of the intermolecular forces in these substances via the magnitude of the temperature change. You will encounter two types of organic compounds in this experiment—alkanes and alcohols. The alkanes you will be investigating are n-pentane (C_5H_{12}) and n-hexane (C_6H_{14}). Notice that the alkanes contain only carbon and hydrogen. The alcohols that you will be studying are methanol (CH_3OH), ethanol (C_2H_5OH), 1-propanol (C_3H_7OH), and 1-butanol (C_4H_9OH); note that in addition to carbon and hydrogen atoms, alcohols contain the $-OH$ functional group. Keep this functional group in mind when drawing the Structural Formulas of these molecules.

You will examine the molecular structure of these alkanes and alcohols for the presence and relative strength of two intermolecular forces—specifically, **hydrogen bonding and dispersion forces**.

PRE-LAB EXERCISE

Prior to doing the experiment, complete the Pre-Lab Table (Table 1, page 5), which gives the name and chemical formula for each compound. First, draw the structural formula and determine the molar mass for each compound. Next, state whether hydrogen bonding exists in a sample of each compound; note that hydrogen bonding occurs when a hydrogen atom is bonded directly to a N, O, or F atom within the molecule. Although the table does not contain a heading for dispersion forces, keep in mind that this type of force exists among all molecules, and generally increases as the molar mass of the molecule increases.

PROCEDURE

ALL OF YOUR DATA and DATA TABLES SHOULD BE DIRECTLY ENTERED INTO THE DATA SECTION OF YOUR LABORATORY NOTEBOOK.

1. Wear safety goggles! **CAUTION:** The compounds used in this experiment are flammable and poisonous. Avoid inhaling their vapors. Avoid contacting them with your skin or clothing. Be sure there are no open flames in the lab during this experiment. Notify your instructor immediately if an accident occurs.

2. Connect a temperature probe to a Go!Link adapter and plug the Go!Link into the USB port on the computer. Type "Logger Pro" into the search bar on the desktop from a logged in computer and open your data collection software called Logger Pro. You will see a live read out for the temperature probe in the upper left-hand corner of the Logger Pro window.

3. You are measuring temperature in degrees C on the y-axis and time in minutes on the x-axis. From the top click on Experiment and drop down to click on Data Collection. Change the time-based collection to units of minutes with duration of 5. Adjust the sampling rate to 60 samples/minute. Click done. You can then use this for all data collection today.

4. Wrap the end of the probe with a square piece of filter paper secured by a small rubber band as shown in Figure 1. Roll the filter paper around the probe tip in the shape of a cylinder. Hint: First slip the rubber band up on the probe, wrap the paper around the probe, and then finally slip the rubber band over the wrapped paper. The paper should be even with the probe end.

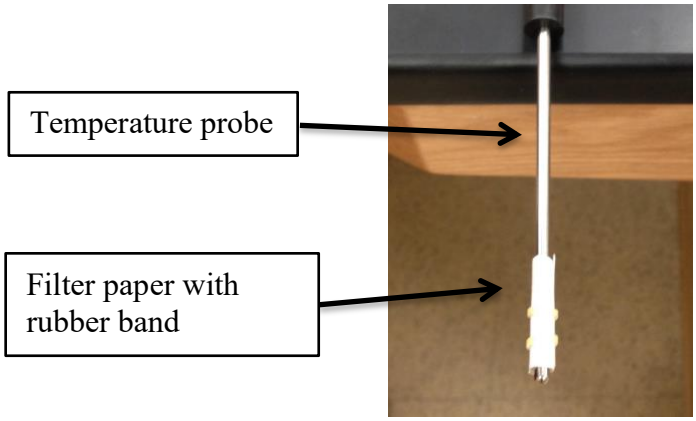


Figure 1: Example of roll of paper around the end of the probe.

5. Place the probe in the **ethanol** container. Make sure the container does not tip over.
6. Prepare a piece of tape, about 10 cm long, to be used to tape the probe in position during the collection process.
7. After the probe has been in the liquid container for about 15 seconds, begin data collection by clicking **COLLECT** or hitting the space bar. Monitor the temperature while the probe is in the container for another 15 seconds to establish a stable initial temperature of the liquid. Remove the probe from the liquid container and tape it to the benchtop so the probe tip extends about 5 cm over the edge of the bench top as shown in Figure 2.

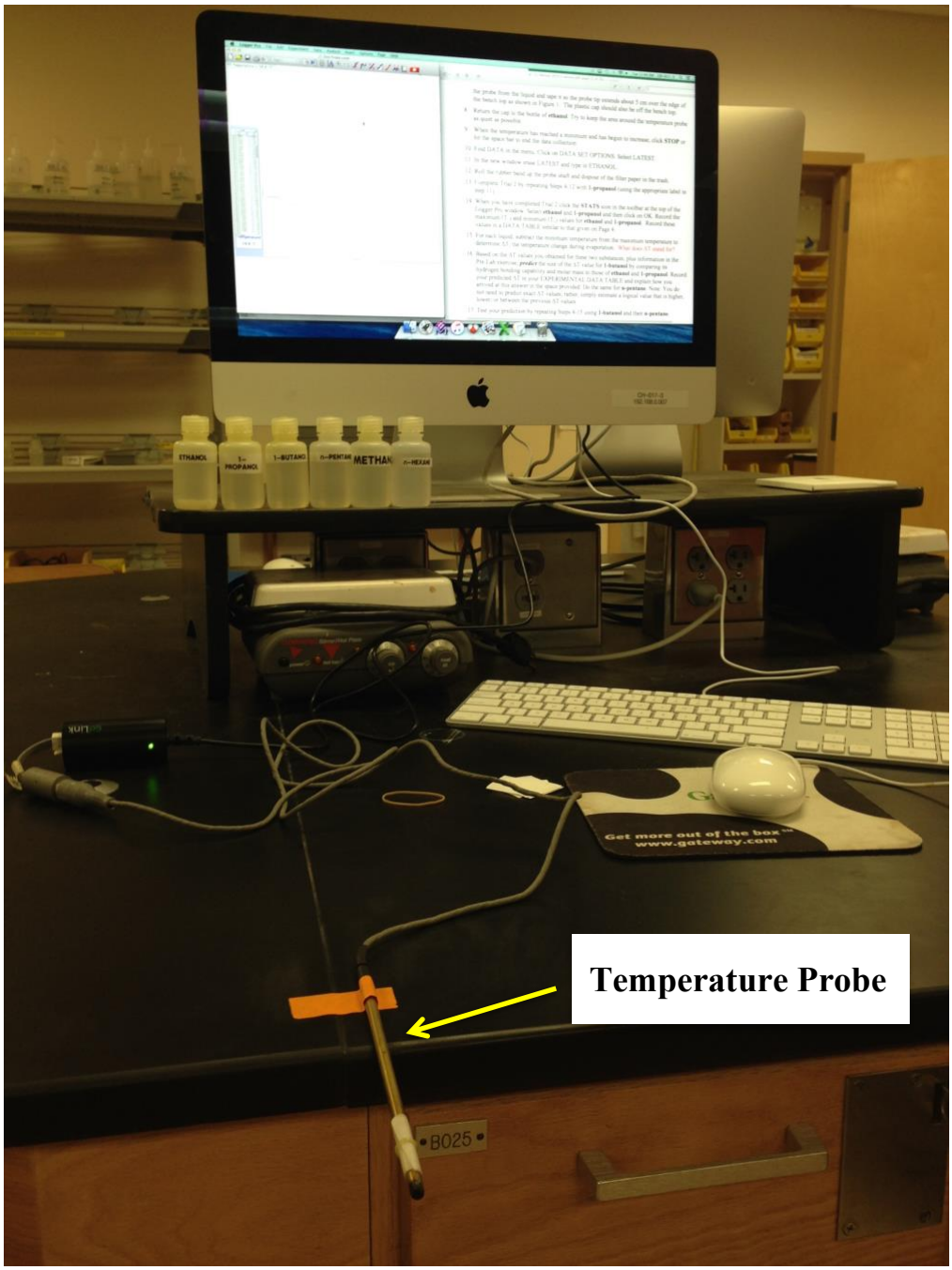


Figure 1: Example of the temperature probe set up

8. Return the cap to the container of **ethanol**. Try to keep the area around the temperature probe as quiet as possible.
9. When the temperature has reached a minimum and has begun to increase, click **STOP** or hit the space bar to end the data collection.
10. Find DATA in the menu. Click on DATA SET OPTIONS. Select LATEST.
11. In the new window, erase LATEST and type in ETHANOL.

12. Roll the rubber band up the probe shaft and dispose of the filter paper in the trash. Save the rubber bands.
13. Complete Trial 2 by repeating steps 4-12 with **1-propanol** (using the appropriate label in step 11).
14. When you have completed Trial 2 click the **STATS** icon in the toolbar at the top of the Logger Pro window. Select **ethanol** and **1-propanol** and then click on OK. Record the maximum (T_1) and minimum (T_2) values for **ethanol** and **1-propanol**. Record these values in Table 2, shown on the next page.
15. For each liquid, subtract the minimum temperature (T_2) from the maximum temperature (T_1) to determine ΔT, the temperature change during evaporation.
16. Based on the ΔT values you obtained for these two substances, plus the information in the Pre-Lab exercise, ***predict*** the size of the ΔT value for **1-butanol** by comparing its hydrogen bonding capability and molar mass to those of **ethanol** and **1-propanol**. Record you predicted ΔT in your EXPERIMENTAL DATA TABLE and explain how you arrived at this answer in the space provided. Do the same for **n-pentane.**

    Note: You do not need to predict the exact ΔT values; rather, simple estimate a logical value that is higher, lower, or in between the previous ΔT values.
17. Test your prediction by repeating Steps 4–15 using **1-butanol** and then **n-pentane**.
18. Based on the ΔT values you have obtained for all four substances, plus information in the Pre-Lab exercise, predict the ΔT values for **methanol** and **n-hexane**. Compare the hydrogen-bonding capability and molar mass of **methanol** and **n-hexane** to those of the previous four liquids. Record your predicted ΔT, and then explain how you arrived at this answer in the space provided.
19. Test your prediction in Step 18 by repeating Steps 4-15, using **methanol** and **n-hexane.**
20. Create a plot by graphing of ΔT values of the **four alcohols** versus their respective molar masses. (We recommend using Excel. Please ask your instructor or TA for assistance if you are not familiar with using Excel or a similar graphing program.) Don't graph the alkanes. Your graph should have a proper title, appropriately sized axes with labels and units and a linear trend line. This graph should be included it in your data section of your submitted lab notebook.

WASTE AND CLEAN UP

1. The rubber bands can be reused, please return them to the appropriate tub.
2. The filter paper can go in the trash.
3. Separate the temperature probe from the Go!Link and the adapter and return them to the appropriate tubs.
4. Quit Logger Pro. Throw away all files that you have saved to the desktop. Shut down your computer.

ADDRESS THE FOLLOWING QUESTIONS IN YOUR CONCLUSION

1. n-pentane and 1-butanol have very similar molar masses, but significantly different ΔT values. Based on their intermolecular forces, explain the difference in ΔT values of these substances.
2. Which of the alcohols studied has the strongest intermolecular forces of attraction? The weakest intermolecular forces? Explain using the results of this experiment using the plot you created.
3. Which of the alkanes studied has the stronger intermolecular forces of attraction? The weaker intermolecular forces? Explain using the results of this experiment.

INCLUDE THE FOLLOWING TABLES IN YOUR DATA SECTION

Table 1. Pre-Lab Table

Substance	**Formula**	**Structural Formulas**	**Molar Mass** $(\frac{g}{mol})$	**Hydrogen-Bond? (Yes or No)**
ethanol	C_2H_5OH			
1-propanol	C_3H_7OH			
1-butanol	C_4H_9OH			
n-pentane	C_5H_{12}			
methanol	CH_3OH			
n-hexane	C_6H_{14}			

Table 2. Experiment Data Table

Substance	$\boldsymbol{T_1}$ (°C)	$\boldsymbol{T_2}$ (°C)	$\boldsymbol{\Delta T(T_1 - T_2)}$ (°C)	**Predicted** $\boldsymbol{\Delta T}$ (°C)	**Explanation**
ethanol				NA	NA
1-propanol				NA	NA
1-butanol					
n-pentane					
methanol					
n-hexane					